Robbie Hao

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SUMMARY

M.S. Statistics candidate with applied machine learning experience building scalable predictive pipelines and evaluation workflows on large datasets. Developed Python factor research scripts, and analytical reports, evaluating 101 alpha signals across ~5,000 equities and 12 years of market data. Skilled in statistical modeling, time-series analysis, and factor research, with experience working with financial data to evaluate signals and portfolio performance.

EDUCATION

Master of Statistical Science—Duke University Aug. 2025–Expected May 2027

B.A. in Economics and Statistics—DUFE Sept. 2021–Jul. 2025

WORK EXPERIENCE

Founder Fubon Fund

Quantitative Researcher Intern Jan.–Mar. 2024; Jul.–Sept. 2024

- Built Python scripts to implement and test 101 Alpha-style factors across ~5,000 equities and 12 years of market data, supporting systematic signal evaluation and factor research.
- Screened and compared candidate factors using IC/ICIR analysis, rank-based portfolio tests, correlation checks, Lasso, and Random Forest selection to identify stronger and less redundant signals.
- Prepared research reports for senior analysts summarizing factor performance, signal stability, risk-return behavior, turnover, and out-of-sample results.
- Applied econometric frameworks (Fama-MacBeth, Fama-French) to evaluate statistical significance, exposure overlap, and robustness of factor construction.

RESEARCH

Graduate Researcher—Duke | Reinforcement Learning for Optimal Execution April 2026 – Present

- Built a DQN execution agent in a multi-agent limit-order-book simulator (ABIDES); designed a 5-dim observable cost state (rolling spread, order-book imbalance, price impact, realized slippage, top-5 book depth) enabling online regime inference, and calibrated liquidity/volatility regimes to empirical spread and VIX anchors
- Diagnosed reward gaming in which agents under-executed 4–7% of parent orders to exploit a linear completion penalty, distorting cross-regime cost comparisons by up to 30%; replaced the penalty with live-book terminal settlement using exact reward folding ($\gamma=1$) and validated backward compatibility with regression tests.
- Ran a cross-regime transfer study across 30 trained policies (3 regimes × 5 seeds × 2 reward designs; 55k+ held-out evaluation episodes), finding asymmetric failure concentrated on timing risk: volatility-trained policies incurred ~2× execution cost in calm markets, while reverse transfer was nearly costless.

Research Assistant—The Second Hospital of Dalian Medical University May 2024−May 2025

- Built an ML feature-selection and prediction pipeline for high-dimensional lipidomics data (498 features, 140 samples), identifying 10 biomarkers with ~0.95 diagnostic AUC.

LEADERSHIP & SCHOOL ACTIVITIES

President, Photography Association—DUFE 2023–2024

- Partnered with Nikon to lead a photography competition with ~300 participants, highlighting hometown culture and raising funds for underserved communities.

TECHNICAL SKILLS

Programming: Python (Pandas, NumPy, SciPy, Matplotlib, PyTorch), R, SQL.

ML/Stats: Random Forest, LightGBM, XGBoost, Lasso/Ridge Regression, Scikit-Learn, TensorFlow, Deep Learning methods, Reinforcement Learning, Bayesian methods

Finance: multi-factor equity models, backtesting, alpha research

Data & Tools: Git, Linux, SLURM (HPC cluster computing), LaTeX, Excel